

Sec 2.5 – The Second Derivative

(these notes are adapted from a variety of sources, including notes provided by Dr. Kristin Camenga)

First, let's discuss some notation.

Given a function $f(x)$, we have written its derivative as $f'(x)$, and we say " f prime." We can also use Leibniz notation for the derivative, $\frac{d}{dx}f(x)$ or $\frac{df}{dx}$.

Next, since the derivative is itself a function, we can consider its derivative.

For a function f , the derivative of its derivative is called the **second derivative**, and written $f''(x)$, and we say " f double-prime."

In Leibniz notation, if $y = f(x)$, the second derivative is written as $\frac{d^2y}{dx^2}$.

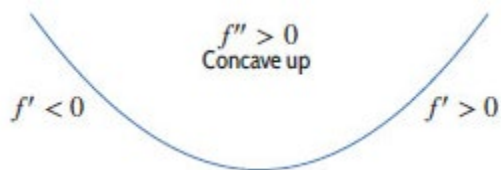
Recall that the derivative of a function tells you whether a function is increasing or decreasing:

- If $f' > 0$ on an interval, then f is *increasing* over that interval.
- If $f' < 0$ on an interval, then f is *decreasing* over that interval.

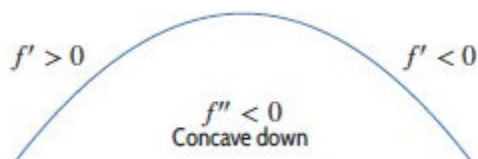
Since f'' is the derivative of f' ,

- If $f'' > 0$ on an interval, then f' is *increasing* over that interval.
- If $f'' < 0$ on an interval, then f' is *decreasing* over that interval.

When f' is increasing, the curve is bending upward, and we say that the original function f is **concave-up**. In the curve below, on the left, the slopes/derivative are getting less negative, so they **are** increasing. It's easy to see that the slopes/derivatives on the right are increasing, or getting steeper.

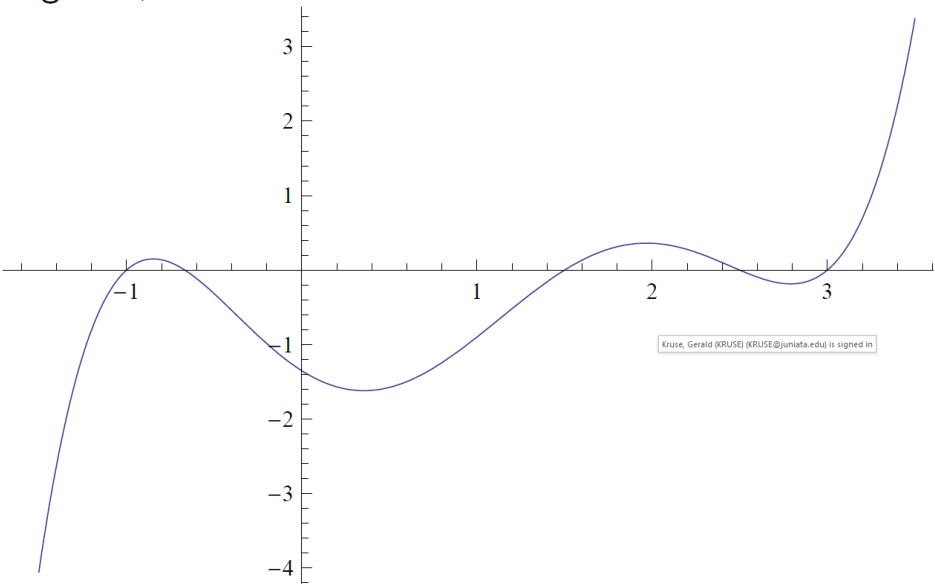


When f' is decreasing, the curve is bending downward, and we say that the original function f is **concave-down**. In the curve below, on the left, the slopes/derivative are getting less positive, so they are decreasing. The slopes/derivatives on the right are decreasing since they are getting more negative.



Partner up

Without sketching, where is the second derivative of this function positive, negative, or zero?



Fill in the blanks:

f	f'	f''
increasing		
	negative	
	0	
		positive
concave down		
		0

So we have

f	f'	f''
increasing	f' is positive, $f' > 0$	don't <u>know</u>
decreasing	negative $f' < 0$	don't <u>know</u>
steady, flat	0	?
f concave UP	f' <u>increasing</u>	positive, $f'' > 0$
concave down	f' <u>decreasing</u>	negative, $f'' < 0$
linear	steady, flat	0

Consider this Garfield cartoon:

Garfield by Jim Davis for July 24, 1992



How can this be interpreted using first and second derivatives?

Solution: We consider the function $f(t)$ which measures Garfield's weight as a function of time. Since Garfield is gaining weight, $f'(t) > 0$. Since the weight gain is slowing down, $f''(t) < 0$.

2.5 #24 Let $P(t)$ represent the price of a share of stock of a corporation at time t . What does each of the following statements tell us about the signs of the first and second derivatives of $P(t)$.

- ① "The price of the stock is rising faster and faster."
- ② "The price of the stock is close to bottoming out."

SOLUTIONS:

- ① Since the price is rising, $P'(t) > 0$. Since it is rising faster and faster, $P''(t) > 0$.
- ② This suggests the price is dropping, but cannot drop much further. Since the price is dropping, $P'(t) < 0$. Since it is bottoming out, the drop is getting smaller and smaller and $P''(t) > 0$ (the rate of change is increasing since it is less negative).

The second derivative measures the rate of change of the derivative.

For example, if $s(t)$ is a position function, with position measured in meters and time in seconds, we know that its derivative $s'(t) = v(t)$ is the velocity with units of meters per second. Velocity tells us how the position is changing.

Then the second derivative, $s''(t) = v'(t) = a(t)$ tells us how the velocity is changing. This is called *acceleration* and the units are (meters per second) per second or meters per second squared. Repeating the units calculation algebraically:

$$\frac{m/s}{s} = \frac{m}{s} \cdot \frac{1}{s} = \frac{m}{s^2}.$$